ELEC5305 Project Proposal

1 Project Title

Handcrafted Features versus Log-Mel CNNs for Environmental Sound Classification under Additive Noise

2 Student Information

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3 Project Overview

Environmental sound classification labels short recordings as events such as dog bark, rain, siren, or breaking glass. Background noise can make clean benchmark results misleading in practical monitoring and assistive devices.

This project compares two pipelines: a support vector machine (SVM) trained on handcrafted temporal, spectral, and cepstral features, and a compact convolutional neural network (CNN) operating on log-Mel spectrograms. Both systems will be evaluated on the same clean and noise-corrupted recordings. The engineering research question is:

> **How do handcrafted audio features and log-Mel spectrogram representations compare in environmental sound classification accuracy, robustness to additive noise, and computational cost?**

Three hypotheses will be tested. First, the CNN will achieve a higher clean-condition macro F1 score because it can learn local time-frequency patterns. Second, both systems will degrade at low signal-to-noise ratio (SNR). Broadband noise can alter the CNN's full spectrogram input and systematically shift handcrafted descriptors such as spectral centroid and roll-off. The CNN's clean-condition advantage is expected to narrow as the SNR decreases, although the experiment will determine which representation degrades more gracefully. Third, noise augmentation will reduce the relative F1 loss of both systems.

4 Background and Motivation

ESC-50 contains 2,000 five-second, 44.1 kHz mono recordings from 50 balanced classes, organised into five predefined folds [1]. Published results include 39.6% accuracy for an MFCC and zero-crossing-rate SVM, 64.5% for Piczak's CNN, and 81.3% for human listeners [1, 2].

Mel-frequency cepstral coefficients (MFCCs) describe short-term spectral shape, while temporal and spectral descriptors capture energy, brightness, noisiness, and spectral change [3, 4].

Aggregating these measurements produces fixed-length vectors suitable for SVM classification [5]. CNNs take a different approach by learning local patterns directly from time-frequency representations [2]. Audio augmentation can improve CNN generalisation when labelled data are limited [6], and a recent review surveys both handcrafted and learned approaches [7]. Because published studies often use different models and protocols, a controlled comparison using the same mixtures, folds, and hardware can provide a clearer engineering trade-off between robustness, accuracy, and cost.

I chose this topic because it applies the Week 3 feature-analysis and Week 4 short-time Fourier transform (STFT) material to a practical robustness problem for resource-constrained monitoring devices.

5 Proposed Methodology

The project will use Python, NumPy, SciPy, scikit-learn, and PyTorch. Custom audio signal processing code will be developed for framing, windowing, STFT computation, feature aggregation, and target-SNR mixing. The custom STFT implementation will be validated against SciPy and the course `MyStft.m` implementation.

Audio will be processed at 44.1 kHz using a 1,024-sample Hann window (23.2 ms), a 512-sample hop (11.6 ms), and a 1,024-point fast Fourier transform (FFT). This configuration gives approximately 43.1 Hz frequency spacing while retaining the time resolution needed for transient sounds. The learned pipeline uses 64 Mel bands; the handcrafted pipeline instead extracts root-mean-square (RMS) energy, zero-crossing rate, spectral centroid, spread, roll-off, flux, 20 MFCCs, and first-order MFCC deltas. Frame-level means and standard deviations then form one feature vector per clip for SVM classification. A compact stack of convolution, normalisation, pooling, and dropout layers operates on the log-Mel spectrograms.

ESC-50's official folds define the outer evaluation. Within each outer split, one of the four non-test folds serves as the validation set for model selection and early stopping. After the settings have been selected, the model is retrained on all four non-test folds. The CNN will be trained with three fixed seeds per outer fold, with metrics averaged across seeds before aggregation across folds. Test data remain excluded from standardisation and model selection.

The primary experiment will train on clean audio and test at clean, 10 dB, 5 dB, and 0 dB. Evaluation noise will include additive white Gaussian noise and channel 1 from held-out DEMAND recordings, including traffic, metro, and station environments [8]. Noise will be scaled to reach the target SNR, defined as

$$\mathrm{SNR}_{\mathrm{dB}} = 10 \log_{10}\left(\frac{P_{\mathrm{signal}}}{P_{\mathrm{noise}}}\right). \tag{1}$$

Signal power will be measured over active frames of the clean clip using a fixed threshold 40 dB below its peak frame energy. DEMAND noise will be resampled to 44.1 kHz before mixing. A manifest will record the noise file, channel, offset, gain, and random seed so both systems receive exactly the same mixtures. In the augmentation extension, both systems will be retrained with noisy examples. DEMAND environments reserved for evaluation will be excluded from augmentation. Because DEMAND recordings may contain engines, crowds, or transport sounds resembling ESC-50 classes, possible label conflicts will be examined per class.

Results will report the mean and standard deviation of accuracy and macro F1 across the five outer folds, together with confusion matrices and relative macro-F1 loss from clean to each noisy condition. Paired bootstrap confidence intervals will compare the two systems on the same test clips. Computational cost will be measured on one CPU with batch size one using feature-processing time, median inference time over 100 runs after warm-up, parameter count, and model-file size.

6 Expected Outcomes

The project will produce two complete classifiers and compare their accuracy, noise robustness, and computational cost across four evaluation conditions. The results should indicate whether the CNN's clean-data advantage persists under noise and justifies its computational cost. A command-line demonstration will classify a supplied audio file.

The public repository and GitHub Pages site will provide working code, setup instructions, key results, the written report and video report.

7 Timeline

Weeks	Planned work
1–2	Consider Project Topic.
3–5	Literature review and dataset collection.
6–7	Implement and cross-check the STFT, features, and SNR-controlled mixing.
8–9	Tune, train, and evaluate the SVM and compact CNN using the nested fold procedure.
10–11	Run robustness, paired statistical, error, computational-cost, and noise-augmentation experiments.
12–13	Finalise reproducible code, report, GitHub Pages site, figures, demonstration, and video.

8 References

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[8] J. Thiemann, N. Ito, and E. Vincent, "The Diverse Environments Multi-Channel Acoustic Noise Database (DEMAND): A Database of Multichannel Environmental Noise Recordings," *Proceedings of Meetings on Acoustics*, vol. 19, no. 1, 035081, 2013. doi:10.1121/1.4799597.

A Experimental Design Summary

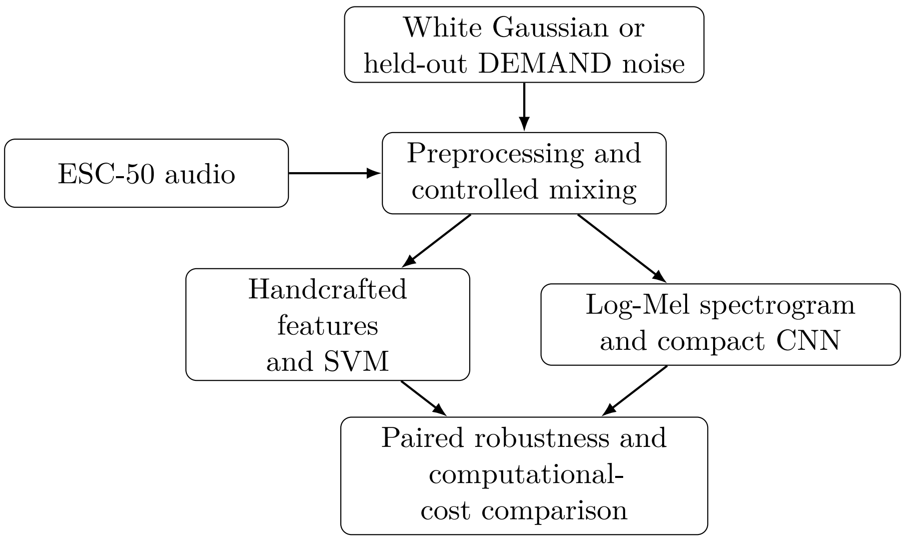

Figure 1: Controlled comparison of the two classification pipelines.

Table 1: Summary of the planned experimental design.

Design element	**Planned conditions**
Systems	Handcrafted features with SVM; log-Mel spectrogram with CNN
Training	Clean primary experiment; matched augmentation extension for both systems
Test noise	Clean, white Gaussian, and held-out DEMAND environments
SNR	Clean, 10 dB, 5 dB, and 0 dB
Validation	Five outer folds, inner validation, and three CNN seeds